

Intermediate Definitions

Note: The second online quiz (in part) is based on these definitions.

Andromeda

Also known by its catalog number M31, the Andromeda galaxy is a large spiral galaxy very similar in appearance to, and slightly larger than, our own Galaxy (Milky Way). At a distance of about 2.5 million light-years, Andromeda is the spiral galaxy that is nearest to our own in space.

Asterism

Any pattern of stars that is not officially recognized by the International Astronomical Union as a **constellation**.

Asteroid Belt

The region of the solar system between the orbits of Mars and Jupiter in which most asteroids are located; the main belt, where the orbits are generally the most stable, extends from 2.2 to 3.3 AU from the Sun

Big Bang

The theory of cosmology in which the expansion of the universe began with a primeval explosion (of space, time, matter, and energy)

Black Hole

A region in spacetime where gravity is so strong that nothing—not even light—can escape

Celestial Sphere

The apparent sphere of the sky; a sphere of large radius centered on the observer; directions of objects in the sky can be denoted by their position on the celestial sphere, and the objects are often referred to as celestial objects

Cosmology

The study of the organization and evolution of the universe or cosmos

Dwarf Galaxy

A small galaxy composed of about 100 million up to several billion stars

Eclipse

Alignment of one celestial body and another that passes between it and the observer.

Ecliptic

The visible path of the Sun as seen against the stars. It is also the projection of Earth's orbit on the sky.

Elliptical Galaxy

A galaxy whose shape is an ellipse and that contains no conspicuous (easily visible) interstellar material

Galaxy Cluster

A structure that consists of anywhere from hundreds to thousands of galaxies that are bound together by gravity

Globular Cluster

One of about 150 large, spherical star clusters (each with hundreds of thousands of stars) that form a system of clusters in the center of our Galaxy; usually has from 10,000 to about a million stars

Hubble Space Telescope

A space telescope that was launched into low Earth orbit in 1990 and remains in operation

Irregular Galaxy

A galaxy without any clear symmetry or pattern; neither a spiral nor an elliptical galaxy

Nova

An existing star that suddenly increases its brightness by many magnitudes — often more than 10 — and then slowly fades.

Nucleus (Galactic)

The round, central, bulging part of the Milky Way or a similar galaxy; in most large galaxies, contains a supermassive black hole

Open Cluster

A comparatively loose cluster of stars, containing from a few dozen to a few thousand members, located in the spiral arms or disk of our Galaxy; sometimes referred to as a galactic cluster

Planetary Nebula

A shell of gas ejected by and expanding away from an extremely hot low-mass star that is nearing the end of its life (the nebulae glow because of the ultra-violet energy of the central star)

Solar Nebula

The cloud of gas and dust from which the solar system formed

Solar Wind

A stream of electrically charged particles emitted from the Sun which travel into space along lines of magnetism.

Spiral Galaxy

A flattened, rotating galaxy with pinwheel-like arms of interstellar material and young stars, winding out from its central bulge

Spiral Arm

A spiral-shaped region, characterized by relatively dense interstellar material and young stars, that is observed in the disks of spiral galaxies

Star Cluster

Group of stars that is very close together in space, held together by gravity, often moving around a common center; usually has from 50 to about a million stars

Supercluster

A large region of space (more than 100 million light-years across) where groups and clusters of galaxies are more concentrated; a cluster of clusters of galaxies

Supernova

An exploding star.

Thermonuclear

Of or related to nuclear fusion reactions

Transit

The visible journey of a planet across the Sun's disc or of a planet's moon across the disc of that planet.
Compare with eclipse.

Transparency

The degree to which light from outside is able to pass through Earth's atmosphere.

Zodiac

The visible zone where the Sun, the Moon and most planets are seen, lying within 10 degrees of the celestial equator. Also, it is the twelve constellations (Aries, Taurus, Gemini, Cancer, Leo, Virgo, Libra, Scorpio, Sagittarius, Capricorn, Aquarius and Pisces) that formed an ancient calendar of the Sun's progress in the sky during one year.